THE MASTERMIND COACHING CLASSES

CBSE / RBSE · Nursery – Class XII · Road No. 5, VKIA, Jaipur · 9887258679

CLASS IX — SCIENCE (086) · MID-TERM / HALF-YEARLY PRACTICE PAPER

Session 2026-27 · School Examination Pattern

Class: 9	Subject: Science	Set 02	Time: 3 Hours	Max. Marks: 80

General Instructions:

1. This question paper consists of 39 questions in 3 sections. Section A is Biology (Q.1–16), Section B is Chemistry (Q.17–27) and Section C is Physics (Q.28–39).
2. All sections are compulsory. However, internal choice is provided in some questions. A student is expected to attempt only one of the options in such questions.
3. Section A carries 32 marks, Section B 25 marks and Section C 23 marks. Marks are printed against each question.
4. Draw neat, labelled diagrams wherever required. Use g = 10 m s^{-2} wherever needed.

SECTION A — BIOLOGY — 16 questions · 32 marks

Q.1–7 MCQs (1 mark each); Q.8–9 Assertion–Reason. For Assertion–Reason: (a) Both A and R are true, and R is the correct explanation of A. (b) Both A and R are true, but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.

1. Which of the following correctly matches a cell organelle with its function? **1**

(a) Ribosome – digestion of worn-out organelles
(b) Lysosome – synthesis of proteins
(c) Golgi apparatus – packaging and dispatch of materials
(d) Mitochondria – photosynthesis

2. A wilted lettuce leaf becomes crisp when it is kept in tap water. This is because: **1**

(a) minerals diffuse out of the cells
(b) the cell walls dissolve in water
(c) water leaves the cells by osmosis
(d) water enters the cells by osmosis and makes them turgid

3. Leaves and stems of a canna plant bend easily without breaking. The tissue that provides this flexibility is: **1**

(a) parenchyma (b) collenchyma (c) sclerenchyma (d) xylem

4. Blood is classified as a connective tissue because it: **1**

(a) flows inside blood vessels
(b) has a fluid matrix (plasma) and connects different parts of the body by transporting substances
(c) contains haemoglobin
(d) contains cells

5. Which substances released from old refrigerators, air-conditioners and aerosol sprays damage the ozone layer? **1**

(a) Carbon monoxide (b) Chlorofluorocarbons (CFCs)
(c) Methane (d) Nitrogen

6. Mushrooms and moulds are placed in kingdom Fungi because they: **1**

(a) make food by photosynthesis
(b) are prokaryotes
(c) have chitin in their cell wall and absorb food from dead organic matter
(d) are motile and ingest food

7. Which part of a flower develops into the fruit after fertilisation? **1**

(a) Ovule (b) Ovary (c) Stigma (d) Anther

8. **Assertion (A):** Tendons are strong but have very little elasticity. **1**
Reason (R): Tendons attach muscles to bones.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

9. **Assertion (A):** An increase in the amount of carbon dioxide in the atmosphere raises the temperature of the Earth. **1**
Reason (R): Carbon dioxide absorbs outgoing infrared radiation and re-radiates part of it back to the Earth.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

10. Write any two differences between self-pollination and cross-pollination. **2**

OR

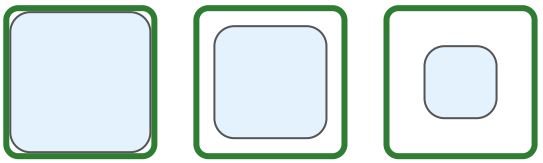
Name two agents of pollination and give one plant pollinated by each.

11. Name the kingdom to which each of the following belongs and write one distinguishing feature: (i) *Amoeba* (ii) mushroom (iii) mango tree. **3**

OR

Differentiate between monocots and dicots on any three points, with one example each.

12. The figure shows the same plant cell kept in three different solutions X, Y and Z. State whether each solution is hypotonic, isotonic or hypertonic to the cell sap. Also name the phenomenon observed in the cell Z. **3**

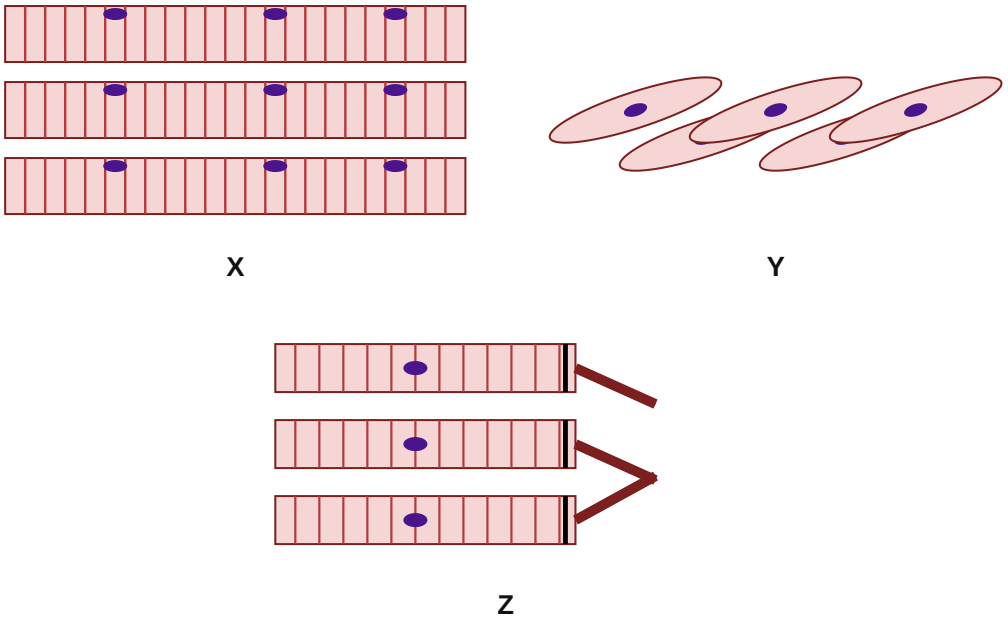


X Y Z

13. Fill in the blanks of the table: **3**

Tissue	Location	Function
Phloem	Vascular bundles of plants	(a) _____
(b) _____	Wall of the heart	Rhythmic contraction to pump blood
Areolar tissue	(c) _____	Supports internal organs and helps in repair

14. The figure shows three types of muscle fibres X, Y and Z. Identify each and name one place in the human body where it is found. **3**



X **Y**

Z

15. While playing football, Ravi twisted his ankle badly. The doctor said that a ligament was torn and advised rest. Ravi's teacher later explained that ligaments help the joints work, and that the human body has different types of joints such as those at the knee, shoulder and neck.

(a) What is a ligament? State its function. 1

(b) Name the type of joint present at the knee. 1

(c) Differentiate between a tendon and a ligament (any two points). 2

OR

(c) Name the type of joint at the (i) shoulder and (ii) neck (skull–vertebra). State one movement each permits.

16. (i) State the three main postulates of the cell theory. **5**
(ii) Name the scientist who first observed cells and the material he used.

OR

Distinguish between a plant cell and an animal cell on any five points.

SECTION B — CHEMISTRY — 11 questions · 25 marks

Q.17–19 MCQs; Q.20 Assertion–Reason (options as in Section A).

17. A mixture of acetone (boiling point 56 °C) and water (boiling point 100 °C) can best be separated by: **1**

(a) filtration (b) simple distillation
(c) using a separating funnel (d) centrifugation

18. The percentage by mass of oxygen in calcium carbonate, $CaCO_3$ (Ca = 40, C = 12, O = 16), is: **1**

(a) 12 % (b) 40 % (c) 48 % (d) 60 %

19. Which of the following best describes a scientific model? **1**

(a) An exact full-size copy of a system
(b) A simplified representation used to explain or predict the behaviour of a system
(c) A final, never-changing truth
(d) A guess that is never tested

20. **Assertion (A):** Colloidal particles can be seen with the naked eye. **1**
Reason (R): Colloids scatter a beam of light passing through them.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

21. Why is a mixture of salt and water separated by evaporation, whereas a mixture of two miscible liquids is separated by distillation? **2**

22. 4 g of hydrogen combines completely with 32 g of oxygen to form water. State the law and find the mass of water formed. **2**

23. Write the electronic configuration of sodium (atomic number 11). State its valency and give a reason. **2**

24. 25 g of sugar is dissolved in 225 g of water. (i) Calculate the % m/m of the solution. (ii) How much water must be added to 100 g of this solution to reduce its concentration to 5 %? **3**

25. How does fractional distillation differ from simple distillation? Give one example where it is used. **3**

OR

Why is crystallisation preferred to evaporation to dryness for purifying a solid? Give one example.

26. A student prepared three sugar solutions of different concentrations at room temperature, as shown in the table.

Beaker	Sugar (g)	Water (g)
P	5	95
Q	10	90
R	20	80

(Solubility of sugar at room temperature ≈ 200 g per 100 g water.)

(a) Which beaker has the highest concentration (% m/m)? 1

(b) Calculate the % m/m of beaker Q. 1

(c) Is the solution in beaker R saturated or unsaturated? Justify. 2

OR

(c) How much sugar is present in 50 g of the solution of beaker R?

27. A kerosene–water mixture and an acetone–water mixture are to be separated. Name the technique for each, give the principle, and describe the procedure for the first. **5**

OR

(a) Differentiate between a solution, a suspension and a colloid on the basis of particle size, Tyndall effect and stability. (b) Why does milk not settle on standing?

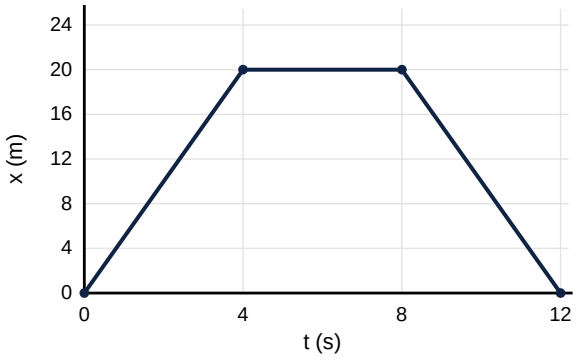
SECTION C — PHYSICS — 12 questions · 23 marks

Q.28–33 MCQs; Q.34 Assertion–Reason (options as in Section A).

28. A cyclist covers the first half of a distance at 10 m s^{-1} and the second half at 15 m s^{-1}. His average speed for the whole journey is: **1**

(a) 12.5 m s^{-1} (b) 12 m s^{-1} (c) 13 m s^{-1} (d) 25 m s^{-1}

29. The displacement–time graph of a body is shown. Its velocity between t = 0 and t = 4 s is: **1**



(a) 5 m s^{-1} (b) 20 m s^{-1} (c) 0 (d) 80 m s^{-1}

30. A 10 kg box is pushed by a horizontal force of 30 N but does not move. The force of friction acting on it is: **1**

(a) zero (b) 10 N (c) 30 N (d) 100 N

31. A truck of mass 2000 kg moving at 5 m s^{-1} and a car of mass 1000 kg moving at 10 m s^{-1} are compared. Their momenta are: **1**

(a) greater for the truck (b) greater for the car
(c) equal (d) cannot be compared

32. A body of mass 1 kg has a kinetic energy of 50 J. Its speed is: **1**

(a) 5 m s^{-1} (b) 10 m s^{-1} (c) 25 m s^{-1} (d) 50 m s^{-1}

33. An echo is heard 2 s after a clap. If the speed of sound in air is 340 m s^{-1}, the distance of the reflecting wall is: **1**

(a) 170 m (b) 340 m (c) 680 m (d) 1360 m

34. **Assertion (A):** The area under a speed–time graph gives the acceleration of the body. **1**
Reason (R): The slope of a velocity–time graph gives the acceleration of the body.

(a) Both A and R are true, and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

35. A constant force of 12 N acts on a 3 kg body initially at rest for 4 s. Find (i) its acceleration and (ii) its velocity at the end of 4 s. **2**

36. A person standing in a bus falls backward when the bus suddenly starts. Name the law that explains this and explain. **2**

OR

Why is it harder to stop a loaded truck than an empty one moving at the same speed?

37. A ship sends an ultrasonic pulse of frequency 30 kHz towards the sea floor and receives the echo after 0.8 s. If the speed of sound in sea water is 1500 m s^{-1}, calculate (i) the depth of the sea and (ii) the wavelength of the pulse. **3**

38. On Sports Day, Rohit tried to push a 40 kg wooden box across the playground. The maximum static friction between the box and the ground is 120 N and the kinetic friction is 80 N.

(a) What is the force of friction on the box when Rohit pushes with 100 N and the box does not move? 1

(b) Rohit now pushes with 160 N and the box starts sliding. Calculate its acceleration. 1

(c) Why is kinetic friction less than maximum static friction? 2

OR

(c) Write two ways in which friction between two surfaces can be reduced.

39. A stone is dropped from the top of a tower and reaches the ground after 4 s ($g = 10\ m\ s^{-2}$). Find (i) its velocity on hitting the ground, (ii) the height of the tower, (iii) the distance fallen in the last second, (iv) the average velocity during the fall and (v) the momentum on hitting the ground if its mass is 0.5 kg. **5**

O R

A car of mass 1000 kg moving at 20 $m\ s^{-1}$ is brought to rest in 10 s by uniform braking. Calculate (i) its initial momentum, (ii) the retardation, (iii) the braking force, (iv) the distance travelled before stopping and (v) the work done by the braking force.